

Does Where You Live Determine *What You Eat*?

Using **actual grocery purchases** by 4.5 million Londoners, this study maps how neighbourhood income shapes diet quality, from protein consumption to drink choices, across every corner of the capital.

Tesco Grocery 1.0 [1]

HMRC Income Data [8]

ONS Household Income [11]

33 Boroughs • 983 MSOAs

33 Boroughs Analysed	983 MSOA Neighbourhoods	$r = .942$ Protein ↔ Health Score	34% Income - Wine Via Age
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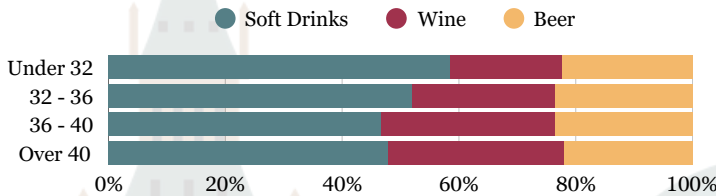
The Question Do richer London neighbourhoods buy higher quality food , more protein, less sugar, and drink differently from poorer ones?	The Approach Grocery spend fractions from Tesco Clubcard data (2015) merged with HMRC [8] and ONS income [11] estimates, analysed at borough and MSOA level.	The Answer Yes, strongly. Income predicts both protein share and drink composition , though age partly explains the wine gradient via residential sorting.
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Insight 1 - Age & Drink Composition

Insight 2 - Income & Protein Intake

As Neighbourhoods Age, *Wine Replaces Soft Drinks*

Drink basket share shifts with average resident age. Wine rises ~35% from youngest to oldest areas, soft drinks fall in near-perfect symmetry. Beer remains age-neutral.[6]



+0.31 Wine r (p<.001)	-0.23 Soft Drinks r	≈0 Beer r (n.s.)
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BUT WHY? MEDIATION ANALYSIS (BARON-KENNY) [2]

Income

a = +0.31 →
p < .001

Wine

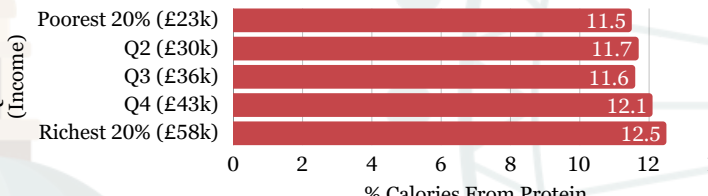
b = +0.20 →
p < .001

Age

Direct path c' = +0.34 (p<.001) · Total effect c = +0.40
Bootstrap indirect (a×b) = **+0.13**, 95% CI excl. zero · ~34% of income–wine gradient operates through age pathway

Richer Boroughs, *Eat More Protein*

Protein-rich foods cost more per calorie than processed carbohydrates. Wealthier boroughs consistently allocate a greater share of basket calories to protein, a direct cost-barrier effect.[4, 5]



⚠ NHS recommended = 20% of calories, no London borough reaches this target

Kensington & Chelsea 12.89% Highest · Affluent Inner West	Newham 10.75% Lowest · Deprived Outer East
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$r = .94$

Protein share ↔ Borough health score
Strongest single finding in the analysis. Protein energy % is an exceptionally reliable proxy for overall dietary quality (p < .001, n = 33 boroughs).

Key assumptions & limitations

Clubcard coverage Only ~44% of Tesco shoppers hold a Clubcard, lower-income and older shoppers are under-represented.	Single retailer Excludes premium (Waitrose, M&S) and discount (Lidl, Aldi) shoppers, true income gradient may be wider.	Ecological fallacy Describes areas, not individuals. LSOA variance is 6× larger than borough, aggregation masks local inequality.	Income data gaps HMRC [8] excludes below-threshold earners; ONS [11] estimates are modelled (CV ≈ 5.7%), not direct observations.
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Finding 1 - Wine & Age (Partially Confirmed)

~34% of the income–wine gradient runs through the age pathway, wealthier areas are older, and older areas buy more wine. Income retains a significant direct effect (c' = +0.34, p < .001) after controlling for age.

Both **purchasing power** and **lifecycle taste** are independent, simultaneous drivers.

Finding 2 - Protein & Income (Confirmed)

Protein energy share is a **strong geographic marker** of dietary quality and income in London. The r = +0.942 correlation with health scores is the strongest finding in this analysis.

A clear **East–West protein** divide mirrors London's deprivation gradient, all boroughs fall below the NHS 20% target.